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Real time dynamic satellite positioning system and positioning method

Abstract

A satellite positioning system with a short initialization processing time and a high fix rate in a Real Time Kinematic satellite positioning system. The Real Time Kinematic satellite positioning system has a mobile station of a user and a transmitting station installed on the ground, and performs positioning using Internet satellites flying above the sky. The mobile station, the transmitting station, and a predetermined Internet satellite are connected to each other by an IPsec-VPN. The positioning system obtains an atomic clock time from the GNSS satellite flying above the sky. The transmitting station obtains its own position information from reference point information obtained from a reference station located within 10 km from the transmitting station and orbit information of predetermined Internet satellites.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a national phase entry under 35 U.S.C. § 371 of International Application No. PCT/JP2022/002546 filed Jan. 25, 2022, which claims the priority from Japanese Patent Application No. 2021-088249 filed in the Japanese Patent Office on May 26, 2021, the entire contents of which are hereby incorporated by reference.

TECHNICAL FIELD

(2) The present invention relates to a positioning system using Real Time Kinematic satellite positioning techniques, and more particularly to a satellite positioning system utilizing Internet satellites.

BACKGROUND

(3) Satellites are divided into a low earth orbit satellite having a global orbit of altitude 2,000 km or less, a mid-earth orbit satellite having a global orbit of altitude 2,000 km to 35,786 km, an Earth synchronous orbit satellite having a global synchronous orbit of altitude 35,786 km, and a high earth orbit satellite having a global orbit of altitude 35,786 km or more. The geosynchronous orbit satellite having a circular orbit over the equator is referred to as a geostationary satellite, but the Earth synchronous orbit satellite may be simply referred to as a geostationary satellite.

(4) Applications of mid-earth orbit satellite includes satellite positioning. By way of example, Global Navigation Satellite System (GNSS) is known. GNSS includes the Global Positioning System (GPS) of the United States Department of Defense, Glonass operated by the Russian Aerospace Forces (Soviet Union), Galileo Positioning System built by the European Union, China's BeiDou Satellite Positioning System, India's NavIcy Navigation Satellite System, etc.

(5) The quasi-zenith satellite system of Japan “MICHIBIKI” is a satellite positioning system consisting of a mid-earth orbit satellite with an altitude of 33,000 to 39,000 km. The satellite has an inclined stationary orbit that draws an 8-shaped North-South symmetry at approximately the top of Japan (quasi-zenith). Japan's technology demonstration satellite “KIZUNA” is an Internet satellite with a geostationary orbit at an altitude of 36,000 km and a longitude of 43 degrees east, but tests of KIZUNA have been completed.

(6) In recent years, satellite Internet access using a satellite constellation has become known as an application of low earth orbit satellites. A satellite constellation consists of many small polar-

orbiting satellites that work together to provide Internet access. In particular, the development of satellite Internet access technology that uses satellite constellations using non-geostationary satellites at an altitude of about 500 km is active.

PRIOR ART LITERATURE

Patent Literature

(7) Patent Document 1: WO 2016/147569

SUMMARY OF THE INVENTION

Problem to be Solved by the Invention

(8) In recent years, Real Time Kinematic GNSS has become known as a high-precision satellite positioning method using GNSS. This is a technology that allows users to accurately determine their position in real time by combining carrier waves from GNSS satellites and location information data from reference points installed on the ground. Reference points include electronic reference points of the Geospatial Information Authority of Japan, reference points independently set by telecommunications carriers, etc.

(9) Factors affecting the accuracy of Real Time Kinematic GNSS positioning include delay error due to ionosphere or troposphere, satellite orbit error, clock error due to errors of a precision clock installed on the satellite or receiver, inaccuracy of the position of the reference point, etc.

(10) Since conventional Real Time Kinematic GNSS positioning uses complex processing to remove various errors, the time and fix rate of the initialization processing was not a satisfactory value.

(11) It is an object of the present invention to provide a satellite positioning system with a short initialization processing time and a high fix rate in a Real Time Kinematic satellite positioning system.

Means for Solving the Problem

(12) According to an embodiment of the present invention, in a Real Time Kinematic satellite positioning system that has a mobile station of a user and a transmitting station installed on the ground, and performs positioning using Internet satellites flying above the sky, the transmitting station may comprise: a signal processing section of transmission system that generates a transmission datagram, the transmission datagram including a transmission data, the transmission data including orbit information of predetermined Internet satellites, location information of the transmitting station and atomic clock time; a baseband section of transmission system that digitally modulates the transmission datagram to generate an intermediate frequency carrier wave; a front end section of transmission system that frequency converts the intermediate frequency carrier wave to generate a high frequency carrier wave; and an antenna of transmission system that multicasts the high frequency carrier wave toward the predetermined Internet satellites; and the mobile station may comprise: an antenna of reception system that receives a high frequency carrier wave, the high frequency carrier wave being multicast from the predetermined Internet satellites; a front end section of reception system that converts the frequency of the high-frequency carrier wave to an intermediate frequency; a baseband section of reception system that digitally demodulates the intermediate frequency carrier wave to generate a received datagram; and a signal processing section of reception system that calculates the position of the mobile station from the received datagram.

(13) According to an embodiment of the invention, in the Real Time Kinematic satellite positioning system, the signal processing section of reception system may comprise: a phase counter that calculates a phase difference of the intermediate frequency carrier waves from the received datagram; a router of reception system that inputs the received datagram and outputs a received IP packet; a decapsulation unit that decapsulates the received IP packet and extracts a received data, the received data including the orbit information of the predetermined Internet satellites, the location information of the transmitting station and the atomic clock time; and a positioning calculator that calculates the position of the mobile station based on the received data and the phase

difference of the received carrier waves.

(14) According to an embodiment of the invention, in the Real Time Kinematic satellite positioning system, the signal processing section of transmission system may comprise: an encapsulation unit that encapsulates the transmission data to generate a transmission IP packet; and a router of transmission system that inputs the transmission IP packet and outputs the transmission datagram.

(15) According to an embodiment of the invention, in the Real Time Kinematic satellite positioning system, the mobile station, the transmitting station, and the predetermined Internet satellites may be connected to each other by IPsec-VPN.

(16) According to an embodiment of the invention, in the Real Time Kinematic satellite positioning system, the atomic clock time may be obtained from a GNSS satellite flying above the sky.

(17) According to an embodiment of the invention, in the Real Time Kinematic satellite positioning system, the predetermined Internet satellites may include at least four Internet satellites and may be non-geostationary low Earth orbit satellites, the non-geostationary low Earth orbit satellites constituting a satellite constellation consisting of a plurality of satellites.

(18) According to an embodiment of the invention, in the Real Time Kinematic satellite positioning system, the digital modulation may be binary phase shift keying (BPSK).

(19) According to an embodiment of the invention, in the Real Time Kinematic satellite positioning system, the transmitting station may obtain its own location information from reference point information and orbit information of the predetermined Internet satellites, the reference point information being obtained from a reference station within 10 km from the transmitting station.

(20) According to an embodiment of the invention, in the Real Time Kinematic satellite positioning system, the orbit information of the predetermined Internet satellites may be orbit information calculated in real time by an orbit tracking station on the ground.

(21) According to an embodiment of the invention, in the Real Time Kinematic satellite positioning system, the frequency band of the carrier waves may be the Ku band.

(22) According to an embodiment of the invention, in a Real Time Kinematic satellite positioning method that has a user's mobile station and a transmitting station installed on the ground, and performs positioning using Internet satellites flying above the sky, the positioning method may comprise: a step of generating, by the transmitting station, a carrier wave including a transmission data, the transmission data including orbit information of predetermined Internet satellites, location information of the transmitting station and atomic clock time, and multicasting, by the transmitting station, the carrier wave toward the predetermined Internet satellites; a step of converting, by the predetermined Internet satellites, the frequency of a carrier wave multicast from the transmitting station from an uplink frequency to a downlink frequency; a step of multicasting, by the predetermined Internet satellites, the carrier wave converted to the downlink frequency to the mobile station; a step of extracting, by the mobile station, a received data including the location information of the transmitting station, the orbit information of the predetermined Internet satellites and the atomic clock time from a carrier wave multicast from the predetermined Internet satellites; a step of calculating, by the mobile station, a phase difference of the carrier waves, the carrier waves being multicast from the predetermined Internet satellites; and a step of calculating, by the mobile station, the position of the mobile station from the received data and the phase difference of the carrier waves.

(23) According to an embodiment of the invention, in the Real Time Kinematic satellite positioning method, the mobile station, the transmitting station, and the predetermined Internet satellites may be connected to each other by IPsec-VPN, and the predetermined Internet satellites may include at least four Internet satellites and may be non-geostationary low Earth orbit satellites, the non-geostationary low Earth orbit satellites constituting a satellite constellation consisting of a plurality of satellites.

(24) According to an embodiment of the invention, in the Real Time Kinematic satellite positioning method, the atomic clock time may be obtained from a GNSS satellite flying above the sky.

(25) According to an embodiment of the invention, in the Real Time Kinematic satellite positioning method, the frequency band of the carrier waves may be the Ku band.

Effect of the Invention

(26) In accordance with the present invention, a satellite positioning system with a short initialization processing time and a high fix rate in a Real Time Kinematic satellite positioning system can be provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1A is a schematic diagram illustrating a configuration example of a Real Time Kinematic satellite positioning system according to the present embodiment.
- (2) FIG. 1B is a schematic diagram illustrating a configuration example of a Real Time Kinematic satellite positioning system according to the present embodiment.
- (3) FIG. 2A is a diagram illustrating an example of a communication protocol used for multicast in the present embodiment.
- (4) FIG. 2B is a diagram illustrating an example of a configuration of a protocol data unit used for multicast in the present embodiment.
- (5) FIG. 3 illustrates an example of encapsulation and decapsulation of data in the present embodiment.
- (6) FIG. 4A is a diagram illustrating a data structure of an IPv6 (Internet Protocol Version 6) header in the present embodiment.
- (7) FIG. 4B is a diagram illustrating an IPv6 payload data structure in the present embodiment.
- (8) FIG. 5A is a diagram illustrating a portion of the operation of the Real Time Kinematic satellite positioning system according to the present embodiment.
- (9) FIG. 5B is a diagram illustrating a portion of the operation of the Real Time Kinematic satellite positioning system according to the present embodiment.
- (10) FIG. 6A is a diagram illustrating a unit line-of-sight vector E and a baseline vector A in differential GPS method, which is the principle of the Real Time Kinematic satellite positioning system according to the present embodiment.
- (11) FIG. 6B is a diagram illustrating an integer bias in differential GPS method, which is the principle of the Real Time Kinematic satellite positioning system according to the present embodiment.
- (12) FIG. 7A is a diagram illustrating a configuration example of a reception system of a mobile station of the Real Time Kinematic satellite positioning system according to the present embodiment.
- (13) FIG. 7B is a diagram illustrating a configuration example of a transmitting station (earth station) of the Real Time Kinematic satellite positioning system according to the present embodiment.
- (14) FIG. 7C is a diagram illustrating a configuration example of an Internet satellite of the Real Time Kinematic satellite positioning system according to the present embodiment.

DETAILED DESCRIPTION OF THE INVENTION

- (15) Embodiments of the present invention will now be described in detail with reference to the accompanying drawings. In the drawings, the same reference numerals are given to the same element, and the overlapped explanation is omitted.
- (16) With reference to FIG. 1A, an example of a Real Time Kinematic satellite positioning system according to the present invention will now be described. The positioning system of the present example comprises a user's mobile station (rover) **10** and a transmitting station (earth station) **20** that multicasts a carrier wave toward Internet satellites **30** flying above the sky. The Real Time

Kinematic satellite positioning method is called differential GPS method (interferometry positioning method) and uses at least four Internet satellites **30**.

(17) The Internet satellites **30** are not limited to any satellite that has the capability to provide an Internet connection. However, in the present embodiment, the Internet satellites **30** preferably constitute a satellite constellation consisting of a number of non-geostationary satellites, and may be low earth orbit satellites or medium earth orbit satellites, and more preferably, non-geostationary low Earth orbit satellites.

(18) In the present embodiment, the carrier wave used for Internet satellite communication is the radio wave of the microwave band. In the present embodiment, any of the L-band (1.6/1.5 GHz), the S-band (2.6/2.5 GHz), the C-band (6/4 GHz), the Ku-band (14/12 GHz) and the Ka-band (30/20 GHz) may be used, but preferably the Ku-band (14/12 GHz) is used. The carrier wave is a modulated wave in which a navigation message and code information (C/A code, P code) are carried on a sine wave. Generally, when the frequency is low, the propagation loss is small, but a wide band cannot be secured. On the other hand, when the frequency is high, a wide band can be secured, but the propagation loss is large.

(19) With reference to FIG. **1B**, an exemplary configuration of a Real Time Kinematic satellite positioning system according to the present invention will be described. According to this embodiment, the reference point of the reference station **40** may be used to determine the location of the transmitting station (earth station) **20**. Hereinafter, the location information of the reference point is simply referred to as the location information of the reference station **40**. According to this embodiment, the distance between the mobile station **10** and the transmitting station **20** is less than or equal to 10 km. The distance between the transmitting station **20** and the reference station **40** is less than or equal to 10 km. Both of the mobile station **10** and the reference station **40** are within 10 km of the periphery of the transmitting station **20**. Further, according to the present embodiment, the atomic clock time provided by the GNSS satellite **50** is used. The reference station **40** obtains an atomic clock time from the GNSS satellite **50** and transmits the atomic clock time to the transmitting station **20**.

(20) In the present embodiment, the mobile station **10**, the transmitting station **20** and the Internet satellites **30** are connected to each other by an IPsec-VPN (Security Architecture for Internet Protocol-Virtual Private Network). Multicast is used for data transmission between the mobile station **10**, the transmitting station **20**, and the Internet satellites **30**. Multicast is a technique in which a sender (source) simultaneously transmits one packet (data) to a particular plurality of recipients. In multicast, routers at network junctions duplicate a single packet and forward the duplicated packets to many receivers. Therefore, the network can be efficiently utilized.

(21) The information required for Real Time Kinematic satellite positioning according to the present invention is three data, that is, the orbit information of the Internet satellites **30**, the location information of the transmitting station **20** and the atomic clock time. The three information will now be described in detail.

(22) First, the orbit information of the Internet satellites **30** will be explained. The orbit information includes a “rough almanac” with an accuracy of several kilometers, a “broadcast almanac (ephemeris)” with an accuracy of about 2 meters, and a “precision almanac” with an accuracy of 10 cm or less.

(23) The “rough almanac” and the “broadcast almanac” are orbit information included in the navigation message transmitted from the Internet satellites **30**. The mobile station **10** and the transmitting station **20** obtain navigation messages from the Internet satellites **30** via IPsec-VPN.

(24) “Precision almanac” is orbit information calculated based on observation signals at orbit tracking stations on the ground. An example is the IPG almanac provided by the international GPS service. The IPG almanac includes a plurality of almanacs such as a super-breaking almanac, a breaking almanac, and a final almanac. These almanacs include those that are obtained in real time but with slightly lower accuracy, and those that are highly accurate but are obtained several days

later.

(25) The orbit information is preferably highly accurate and obtained in real time. If such conditions are met, the orbit information may be a “broadcast almanac”, but may be a “precision almanac”. In the present embodiment, orbit information obtained directly from the Internet satellites **30** may be used, but orbit information obtained from a ground station may be used.

(26) The orbit of a satellite is represented by the position coordinates of the satellite expressed in three-dimensional coordinates of the Earth's centroid system. The Earth's centroid system is a three-dimensional orthogonal coordinate system whose origin is the Earth's center of gravity. As the Earth's centroid system, the International Earth Reference Coordinate System constructed by the International Earth Rotation Observation Project is known.

(27) The location information of the transmitting station **20** will now be described. In the present embodiment, the transmitting station **20** maintains its own accurate and up-to-date location information. The transmitting station **20** may obtain its own location information independently, or may obtain its own location information of the transmitting station **20** based on the location information of the reference point obtained from the reference station **40**. As mentioned above, the mobile station **10** and the reference station **40** are within 10 km from the transmitting station **20**.

(28) Reference points in Japan may be 1,300 or more electronic reference points nationwide established by the Geospatial Information Authority of Japan, or may be reference points independently established by communication companies, etc. In North America, reference points may be obtained by WGS84 (World Geodetic System, 1984) by the U.S. National Geospatial-Intelligence Agency. In Europe, reference points may be obtained from the European Geodetic Data (European Datum 1950).

(29) Finally, the atomic clock time will be described. The GNSS satellite **50** broadcasts an atomic clock time at predetermined time intervals. The GNSS satellite **50** may be any medium earth orbit satellite that provides an atomic clock time, for example, any satellite selected from Global Positioning System (GPS), Glonass, Galileo Global Navigation Satellite Positioning System, BeiDou Satellite Positioning System, NavIcy Navigation Satellite System or the like. The atomic clock time may be a time obtained by a cesium atomic clock based on international atomic time.

(30) The positioning system of the present embodiment uses the atomic clock time provided by the GNSS satellite **50**. In the present embodiment, the mobile station **10**, the transmitting station **20**, and the reference station **40** have synchronized and therefore identical atomic clock times.

Therefore, it is possible to eliminate atomic clock errors caused by the mobile station **10**, transmitting station **20**, and reference station **40** having different atomic clock times.

(31) For example, the reference station **40** transmits the atomic clock time obtained from the GNSS satellite **50** to the transmitting station **20**. The transmitting station **20** transmits the atomic clock time to the mobile station **10** via Internet satellites **30**. Instead of the reference station **40** receiving an atomic clock time from the GNSS satellite **50**, the transmitting station **20** may receive an atomic clock time from the GNSS satellite **50**. In this case, it is not necessary for the reference station **40** to transmit the atomic clock time to the transmitting station **20**.

(32) With reference to FIG. 2A, an example of a communication protocol used for multicast in the present embodiment will be described. As the application layer protocol, DNS (Domain Name System), NTP (Network Time Protocol), SNMP (Simple Network Management Protocol), etc. may be used. The application layer protocol data unit is sometimes called a message. As the transport layer protocol, UDP (User Datagram Protocol) may be used. The transport layer protocol data unit is usually called a segment, but in UDP called a datagram or UDP packet.

(33) As the Internet layer protocol, IPv6 and IPsec (Security Architecture for Internet Protocol) may be used. The protocol data unit of Internet layer is simply called a packet or IP packet. Since IPsec is an Internet layer encryption protocol, confidentiality of packet can be achieved even if the packet is not encrypted in the higher layer, such as transport layer or application layer protocols. Ethernet may be used as the network access layer protocol. The protocol data unit of the network

access layer is called a frame. The method of transmitting IP packets using Ethernet is called IPoE (IP over Ethernet).

(34) Multicast is a communication method that simultaneously transmits one piece of data to multiple parties belonging to a “multicast group” by simply specifying one destination address. IPv4 (Internet Protocol Version 4) uses IGMP (Internet Group Management Protocol) as a multicast group management protocol, while IPv6 uses MLD (Multicast Listener Discovery). MLD is a protocol for managing participants in multicast groups in IPv6, and has a function to search for multicast receivers (listeners).

(35) According to this embodiment, an IPsec-VPN is formed. IPsec-VPN is a virtual private network that uses IPsec. Therefore, even when using a public network such as the Internet, a secure communication network can be realized by implementing IPsec. IPsec includes an authentication header (AH), an encapsulation of a security payload (ESP: Encapsulation Security Payload), and a key exchange protocol (IKE: Internet Key Exchange).

(36) With reference to FIG. 2B, an example of a configuration of a protocol data unit (PDU) in the present embodiment will be described. The datagram is generated by adding a UDP header to the application layer data. The packet is generated by adding an IP header to the datagram. The IP header includes an IP address. The frame is generated by adding a frame header and an FCS (Frame Check Sequence) to the packet. In the Ethernet, the frame header includes a MAC (Media Access Control) address.

(37) With reference to FIG. 3, an example of encapsulation and decapsulation of data in the present embodiment will be described. A datagram **302** is generated from the transmission data **301**, a packet **303** is generated from the datagram **302**, and a frame **304** is generated from the packet **303**. Each such process is referred to as encapsulation. Conversely, packet **303** is retrieved from frame **304**, datagram **302** is retrieved from packet **303**, and data **301** is retrieved from datagram **302**. Each such process is referred to as decapsulation.

(38) With reference to FIG. 4A, a data structure of IPv6 packet header in the present embodiment will be described. The header of the IP packet is read out by the router and is used for routing or sorting packets. The header of the IPv6 packet is a fixed length of 40 bytes. The “version” represents version 6 of IP. The “traffic class” identifies a difference in class of IPv6 packets and a priority. The “flow label” identifies audio and video data to be transmitted in real time. The “payload length” indicates the total number of bytes of the data and the extension header. The “next header” is a protocol number that represents the type of information following the IPv6 header. The “hop limit” limits the maximum number of routers that can pass. Each time a packet passes through the router, the hop limit is subtracted one by one. When the hop limit reaches zero, the packet is discarded and a hop limit excess is returned to the source. The “source address” is the source IP address and the “destination address” is the destination IP address.

(39) With reference to FIG. 4B, a data structure of IPv6 packet payload in the present embodiment will be described. The payload consists of an extension header and data. The AH (Authentication Header) performs authentication of the data and authentication of the source of the packet. The ESP (Encapsulation Security Payload) has a data authentication function, a packet source authentication and encryption function.

(40) The data includes the orbit information of the Internet satellites **30**, the location information of the transmitting station **20**, the atomic clock time and IKE (Internet Key Exchange). IKE is a key exchange protocol that safely exchanges key information. AH, ESP and IKE are IPsec protocols.

(41) With reference to FIGS. 5A and 5B, an example of the operation of a real-time positioning system according to the present invention will be described. It should be noted that the vertical axes attached to the mobile station **10**, the transmitting station **20**, the Internet (IN) satellite **30**, the reference station **40**, and the GNSS satellite **50** are indicative of the order of processings and do not necessarily represent time. For example, the processings at reference station **40** and transmitting station **20** may be performed independently. The processings at the Internet satellites **30** may also

be performed independently of the processings at the reference station **40** and the mobile station **10**.
(42) As shown in FIG. 5A, mobile station **10**, transmitting station **20**, and at least four Internet (IN) satellites **30** are connected by IPsec-VPN. The transmitting station **20** and the reference station **40** are interconnected by a communication network such as the Internet. In step **S101**, the transmitting station **20** obtains its own accurate and up-to-date location information. Although the transmitting station **20** may uniquely obtain its own location information, the transmitting station **20** may obtain its own location information based on the location information of the reference points obtained from the reference station **40**. In steps **S102** and **S103**, the mobile station **10** and the transmitting station **20** obtain orbit information of the Internet satellites **30** via an IPsec-VPN. From the orbit information, the mobile station **10** selects Internet satellites **30** suitable for user positioning from a number of Internet satellites **30** flying above the sky.

(43) In step **S104**, the mobile station **10** generates and encapsulates a positioning request signal. The positioning request signal may include an identification number and password of the mobile station **10**. The mobile station **10** incorporates the encapsulated positioning request signal into a sine wave to generate a carrier wave. In step **S105**, the mobile station **10** multicasts the carrier wave toward at least four Internet satellites **30**. In step **S106**, each Internet satellite **30** converts the frequency of the carrier wave from an uplink frequency to a downlink frequency. For example, if the Ku band (14/12 GHz) is used as the carrier wave, each Internet satellite **30** converts the frequency from 14 GHz to 12 GHz.

(44) In step **S107**, each Internet satellite **30** multicasts the carrier wave. In step **S108**, the transmitting station **20** receives and decapsulates the carrier wave and retrieves the positioning request signal. In step **S109**, the transmitting station **20** transmits the positioning request signal to the reference station **40**. The transmission from the transmitting station **20** to the reference station **40** may use an existing Internet.

(45) The reference station **40** has obtained orbit information of the GNSS satellites **50** in advance. Therefore, the reference station **40** selects in advance a GNSS satellite **50** suitable for receiving an atomic clock time from a plurality of GNSS satellites **50** flying above the sky. At step **S110**, the reference station **40** obtains an atomic clock time from the GNSS satellite **50**. At step **S111**, the reference station **40** transmits the atomic clock time to the transmitting station **20**.

(46) As shown in FIG. 5B, in step **S112**, the transmitting station **20** generates a transmission data including the orbital information of each Internet satellite **30**, the location information of the transmitting station **20** and the atomic clock time obtained from the GNSS satellite **50**. In step **S113**, the transmitting station **20** encapsulates the transmission data. The transmitting station **20** incorporates the encapsulated transmission data into a sine wave to generate a carrier wave. At step **S114**, the transmitting station **20** multicasts the carrier wave toward at least four Internet satellites **30**.

(47) At step **S115**, each Internet satellite **30** converts the frequency of the carrier wave from an uplink frequency to a downlink frequency. For example, if the Ku band (14/12 GHz) is used as the carrier wave, each Internet satellite **30** converts the frequency from 14 GHz to 12 GHz. In step **S116**, each Internet satellite **30** multicasts the carrier wave. In step **S117**, the mobile station **10** receives and decapsulates the carrier wave and retrieves the received data. The received data includes the orbital information of each Internet satellite **30**, the location information of the transmitting station **20** and the atomic clock time.

(48) In step **S118**, the mobile station **10** computes the phase difference of the carrier waves. In step **S119**, the mobile station **10** computes the relative position between the transmitting station **20** and the mobile station **10** and obtains its own current location in real time.

(49) With reference to FIGS. 6A and 6B, the principle of differential GPS method in a real-time positioning system according to the present embodiment will be described. As shown in FIG. 6A, the position of the transmitting station **20** is a reference point O, the position of the mobile station **10** is P, and the position of the Internet satellite **30** is S. As noted above, the differential GPS

method uses at least four Internet satellites **30**, only one of which is shown here.

(50) Let L be the linear distance from the reference point O to the position P of the mobile station **10**, and let the baseline vector A be a vector starting at the reference point O and ending at the position P of the mobile station **10**. The unit vector starting at the reference point O and pointing toward the position S of the Internet satellite **30** is the unit line-of-sight vector E . Let θ be the angle between the unit line-of-sight vector E and the baseline vector A . The position S of the Internet satellite **30** is actually the position of the antenna on the Internet satellite **30**. The position P of the mobile station **10** is actually the position of the antenna on the mobile station **10**.

(51) Let $D1$ be the distance from the position S of the internet satellite **30** to the reference point O , and let $D2$ be the distance from the position S of the internet satellite **30** to the position P of the mobile station **10**. Let the path difference be δD .

(52) $D = D2 - D1$ Formula1

(53) The three-dimensional position coordinates (x, y, z) of the reference point O are obtained by the transmitting station **20** in advance. The three-dimensional position coordinates of the Internet satellites **30** are obtained from the orbit information. That is, the distance $D1$ is calculated from the reference point O and the position S of the Internet satellite **30**. Therefore, if the path difference δD is obtained, the distance $D2$ can be obtained. The position P of the mobile station **10** is on a spherical surface with a radius $D2$ centered on the position S of the Internet satellite **30**. The position P of the mobile station **10** is the intersection of spherical surfaces with a radius $D2$ centered on the position S of the four Internet satellites **30**. Therefore, by solving the simultaneous equations, the position P of the mobile station **10** can be obtained.

(54) As shown in FIG. 6B, the path difference δD is expressed as the product of the wavelength, of the sine wave and the wave number $(N + \Phi)$.

(55) $D = (N + \Phi) \lambda$ Formula2

(56) N is the integer part of the wavenumber and Φ (Phi) is the fractional part of the wavenumber. The fractional portion Φ (Phi) of the wavenumber is greater than 0 and less than 1. The integer part N of the wavenumber is referred to as an integer value bias or an integer value ambiguity. The step of determining the integer value bias is referred to as an initialization process. The real value of the wavenumber (sum of integer and fractional part) is called the float solution, and the integer part of the wavenumber, i.e. the integer value bias, is called the fix solution. The time until a fix solution is obtained is called the initialization time.

(57) In differential GPS method, a “phase” is used instead of a “wavenumber”. In general, as a unit of the phase of the sine wave, the angle (0 to 360 degrees) or radian (0 to 2π) is used, but in the differential GPS method, cycles (0 to 1) are used. For example, when the phase difference of the carrier waves is 5.2, the integer bias is 5 and the fraction is 0.2.

(58) The carrier waves received by the mobile station **10** and the transmitting station **20** include errors due to ionospheric delays and troposphere delays. However, since the path from the Internet satellite **30** to the mobile station **10** and the path from the Internet satellite **30** to the transmitting station **20** are substantially identical, the ionospheric delay and the troposphere delay included in the two carrier waves may be considered identical. Thus, these delay amounts are erased by subtraction in Equation 1, which determines the path difference δD .

(59) Here, the position coordinates of the mobile station **10** were determined using the position of the transmitting station **20** as a reference point O . Therefore, when the reference point O is the position of the reference station **40**, the position of the transmitting station **20** can be determined.

(60) With reference to FIG. 7A, an example of the configuration of the reception system of the mobile station **10** according to this embodiment will be described. The reception system of the mobile station **10** includes an antenna **101**, a front end section **12**, a baseband section **14**, and a signal processing section **16**. Antenna **101** receives carrier waves multicast from Internet satellites **30**. The front end section **12** generates an intermediate frequency carrier wave from the high

frequency carrier wave received by the antenna **101**. The baseband section **14** digitally demodulates the intermediate frequency carrier wave and generates a datagram which is a UDP packet. The signal processing section **16** generates positioning data, that is, the position of the mobile station **10**, from the datagram.

(61) The front end section **12** includes a low noise amplifier **102** and a frequency converter **103**. The low noise amplifier **102** amplifies a weak high frequency carrier wave with low noise. The frequency converter **103** converts a high frequency carrier wave into an intermediate frequency carrier wave. This intermediate frequency carrier wave is sometimes called a beat signal. The baseband section **14** includes a demodulator (modem) **104** and a correlator (not shown). The demodulator **104** digitally demodulates the intermediate frequency carrier wave to generate a digital received signal, i.e. a datagram.

(62) The signal processing section **16** includes a router **105**, a decapsulation unit **106**, a phase counter **107**, and a positioning calculator **108**. The router **105** has a function of selecting a route or sorting packets in the Internet layer. As described with reference to FIG. 4A, the header of an IP packet includes a source address, a destination address, and the like. If the router **105** determines that the destination address is inappropriate, the router **105** discards the packet and notifies the source address of this fact.

(63) The router **105** inputs the datagram from the baseband section **14** and outputs an IP packet. The decapsulation unit **106** decapsulates the IP packet via the router **105** and retrieves the received data. The received data includes the orbit information of the Internet satellites **30**, the location information of the transmitting station **20** and the atomic clock time. The phase counter **107** inputs the datagram from the baseband section **14** and counts the phase difference of the carrier waves. The positioning calculator **108** calculates the location information of the mobile station **10** from the phase difference of the carrier waves and the received data.

(64) Although FIG. 7A shows a configuration example of the reception system of the mobile station **10**, the reception system of the reference station **40** may have a similar configuration.

(65) With reference to FIG. 7B, an example of a configuration of a transmitting station (earth station) according to the present embodiment will be described. The transmitting station **20** includes an antenna **201**, a front end section **22**, a baseband section **24**, and a signal processing section **26**. The antenna **201** multicasts a carrier wave toward Internet satellites **30**. The front end section **22** generates a high frequency carrier wave from an intermediate frequency carrier wave. The baseband section **24** digitally modulates the transmission datagram to generate an intermediate frequency carrier wave. The signal processing section **26** generates a datagram including a transmission data. This transmission data includes the orbit information of the Internet satellites **30**, the location information of the transmitting station **20** and the atomic clock time.

(66) The front end section **22** includes a power amplifier **202** and a frequency converter **203**. The power amplifier **202** amplifies the high frequency carrier wave to a high power level of several hundred watts. The frequency converter **203** converts an intermediate frequency carrier wave into a high frequency carrier wave. The baseband section **24** includes a modulator (modem) **204**. The modulator **204** digitally modulates the datagram to an intermediate frequency carrier wave. In the present embodiment, the digital modulation system may be any phase shift keying system, but is preferably a binary phase shift keying (BPSK) system.

(67) The signal processing section **26** includes a router **205** and an encapsulation unit **206**. The router **205** inputs the IP packet and outputs the datagram. The router **205** refers to the header of the IP packet to determine route selection and whether transfer is possible. The encapsulation unit **206** encapsulates the transmission data and generates a IP packet. The transmission data includes the orbit information of the Internet satellites **30**, the location information of the transmitting station **20** and the atomic clock time.

(68) With reference to FIG. 7C, an exemplary configuration of an Internet satellite **30** according to the present embodiment will be described. The Internet satellite **30** has antennas **301A** and **301B**, a

demultiplexer **302**, a plurality of repeaters **31**, attitude sensors (star trackers), and the like. Each repeater **31** has a low noise amplifier **303**, a frequency converter **304**, a power amplifier **305**, and a router **306**. Here, the receiving antenna **301A** and the transmitting antenna **301B** may be provided, but one antenna for receiving and transmitting may be used.

(69) The demultiplexer **302** demultiplexes the carrier waves received by the receiving antenna **301A** into each frequency, and distributes the demultiplexed waves to the repeaters for each frequency. The low noise amplifier **303** amplifies a weak carrier wave with low noise. The frequency converter **304** converts the frequency of the carrier wave from the uplink frequency to the downlink frequency. For example, if the Ku band (14/12 GHz) is used as the carrier waves, the frequency converter **304** converts the frequency from 14 GHz to 12 GHz. The power amplifier **305** amplifies the power of the carrier wave. Typically, the satellite output power is tens of watts to hundreds of watts. It should be noted that a band filter may be provided between the frequency converter **304** and the power amplifier **305**. The carrier wave from the power amplifier **305** is transmitted to the demultiplexer **302** via the router **306**. The carrier wave is broadcast via the transmitting antenna **301B**.

(70) While the positioning system according to the present embodiment has been described above, these are merely examples and do not limit the scope of the present invention. Additions, deletions, changes, improvements, etc. that can be easily made to this embodiment by those skilled in the art are within the scope of the present invention. The scope of the invention is defined by the following claims.

EXPLANATION OF REFERENCE NUMERALS

(71) **10**_mobile station (rover), **12**_front end section, **14**_baseband section, **16**_signal processing section, **20**_transmitting station (earth station), **22**_front end section, **24**_baseband section, **26**_signal processing section, **30**_Internet (IN) satellite, **31**_repeater, **40**_reference station, **50**_GNSS satellite, **60**_IPsec-VPN, **101**_antenna, **102**_low noise amplifier, **103**_frequency converter, **104**_demodulator (modem), **105**_router, **106**_decapsulation unit, **107**_phase counter, **108**_positioning calculator, **201**_antenna, **202**_power amplifier, **203**_frequency converter, **204**_modulator (modem), **205**_router, **206**_encapsulation unit, **301A**, **301B**_antenna, **302**_demultiplexer, **303**_low noise amplifier, **304**_frequency converter, **305**_power amplifier, **306**_router

Claims

1. A Real Time Kinematic satellite positioning system that has a mobile station of a user and a transmitting station installed on the ground, and performs positioning using Internet satellites flying above the sky, the positioning system using differential GPS method, the transmitting station comprising: a signal processing section of a transmission system that generates a transmission datagram, the transmission datagram including a transmission data, the transmission data including orbit information of predetermined Internet satellites, location information of the transmitting station and atomic clock time; a baseband section of the transmission system that digitally modulates the transmission datagram to generate an intermediate frequency carrier wave; a front end section of the transmission system that frequency converts the intermediate frequency carrier wave to generate a high frequency carrier wave; and an antenna of the transmission system that multicasts the high frequency carrier wave toward the predetermined Internet satellites; and the signal processing section of the transmission system comprising: an encapsulation unit that encapsulates the transmission data to generate a transmission IP packet; and a router of the transmission system that inputs the transmission IP packet and outputs the transmission datagram, the mobile station comprising: an antenna of a reception system that receives a high frequency carrier wave, the high frequency carrier wave being multicast from the predetermined Internet satellites; a front end section of the reception system that converts the frequency of the high

frequency carrier wave to an intermediate frequency; a baseband section of the reception system that digitally demodulates the intermediate frequency carrier wave to generate a received datagram; and a signal processing section of the reception system that calculates the position of the mobile station from the received datagram, the signal processing section of the reception system comprising: a phase counter configured to calculate, for each of the predetermined Internet satellites, a phase difference between the carrier wave multicast by the transmitting station toward the respective predetermined Internet satellite and the carrier wave multicast from the respective predetermined Internet satellite to the mobile station, the phase difference being calculated based on the received datagram, and the phase difference being indicative of a difference between (i) a distance from the respective predetermined Internet satellite to the transmitting station and (ii) a distance from the respective predetermined Internet satellite to the mobile station; a router of the reception system that inputs the received datagram and outputs a received IP packet; a decapsulation unit that decapsulates the received IP packet and extracts a received data, the received data including the orbit information of the predetermined Internet satellites, the location information of the transmitting station and the atomic clock time; and a positioning calculator that calculates the position of the mobile station based on the received data and the phase difference, wherein the mobile station, the transmitting station, and the predetermined Internet satellites are connected to each other by IPsec-VPN, the predetermined Internet satellites including at least four Internet satellites.

2. The Real Time Kinematic satellite positioning system according to claim 1, wherein the atomic clock time is obtained from a GNSS satellite flying above the sky.

3. The Real Time Kinematic satellite positioning system according to claim 1, the predetermined Internet satellites being non-geostationary low Earth orbit satellites, the non-geostationary low Earth orbit satellites constituting a satellite constellation consisting of a plurality of satellites.

4. The Real Time Kinematic satellite positioning system according to claim 1, wherein the digital modulation is binary phase shift keying (BPSK).

5. The Real Time Kinematic satellite positioning system according to claim 1, wherein the transmitting station obtains its own location information from reference point information and orbit information of the predetermined Internet satellites, the reference point information being obtained from a reference station within 10 km from the transmitting station.

6. The Real Time Kinematic satellite positioning system according to claim 1, wherein the positioning system further comprises an orbit tracking station on the ground, and wherein the orbit information of the predetermined Internet satellites is orbit information calculated in real time by the orbit tracking station on the ground.

7. The Real Time Kinematic satellite positioning system according to claim 1, wherein the frequency band of the carrier waves multicast toward the predetermined Internet satellites and the carrier wave multicast from the predetermined Internet satellites is the Ku band.

8. A Real Time Kinematic satellite positioning method that uses a user's mobile station and a transmitting station installed on the ground, and performs positioning using Internet satellites flying above the sky, the positioning method using differential GPS method, the positioning method comprising: a step of capsulating, by the transmitting station, a transmission data including orbit information of predetermined Internet satellites, location information of the transmitting station and atomic clock time to generate a transmission IP packet; a step of generating, via a router of a transmission system, a transmission datagram from the transmission IP packet, and multicasting a carrier wave including the transmission datagram toward the predetermined Internet satellites; a step of converting, by the predetermined Internet satellites, the frequency of a carrier wave multicast from the transmitting station from an uplink frequency to a downlink frequency; a step of multicasting, by the predetermined Internet satellites, the carrier wave converted to the downlink frequency to the mobile station; a step of generating, by the mobile station, a received datagram from the carrier wave multicast from the predetermined Internet satellites, generating, via a router

of a reception system, a received IP packet from the received IP datagram, and decapsulating the received IP packet to extract a received data including the location information of the transmitting station, the orbit information of the predetermined Internet satellites and the atomic clock time; a step of calculating, by the mobile station, for each of the predetermined Internet satellites, a phase difference between the carrier wave multicast by the transmitting station toward the respective predetermined Internet satellite and the carrier wave multicast from the respective predetermined Internet satellite to the mobile station, the phase difference being indicative of a difference between (i) a distance from the respective predetermined Internet satellite to the transmitting station and (ii) a distance from the respective predetermined Internet satellite to the mobile station a step of calculating, by the mobile station, the position of the mobile station from the received data and the phase difference, wherein the mobile station, the transmitting station, and the predetermined Internet satellites are connected to each other by IPsec-VPN, the predetermined Internet satellites including at least four Internet satellites.

9. The Real Time Kinematic satellite positioning method according to claim 8, wherein the predetermined Internet satellites being non-geostationary low Earth orbit satellites, the non-geostationary low Earth orbit satellites constituting a satellite constellation consisting of a plurality of satellites.

10. The Real Time Kinematic satellite positioning method according to claim 8, wherein the atomic clock time is obtained from a GNSS satellite flying above the sky.

11. The Real Time Kinematic satellite positioning method according to claim 8, wherein the frequency band of the carrier waves multicast toward the predetermined Internet satellites and the carrier wave multicast from the predetermined Internet satellites is the Ku band.
